Name_____ Date_____ Period_____

LESSON 3-1 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Understanding Ratios

Maryland Standard 6.RP.A.1 · Learning target – I can describe a relationship between two quantities using ratio language.

REMEMBER

- A **ratio** compares two quantities: 3 to 5, 3 : 5, or 3/5. **Order matters** – follow the order of the words.
- **Part-to-part** compares two groups. **Part-to-whole** compares one group to the total.
- **Simplest form:** divide both terms by their GCF.

WORKED EXAMPLE

A shelf has 10 science books and 15 history books. Write the ratio of science books to all the books in simplest form.

Step 1 Total: 10 + 15 = 25 books

Step 2 Science first: 10 : 25

Step 3 Divide both terms by 5: 2 : 5

Answer **2 : 5 – for every 2 science books, there are 5 books in all**

WORD BANK

ratio – compares two amounts

part-to-part – one group to another group

part-to-whole – one group to the total

simplest form – no number divides both terms

PRACTICE

show your work · label units

1. A snack bag has 6 pretzels and 9 crackers. Write each ratio in simplest form.

Step 1 Say the first group, then the second.

Step 2 Divide both terms by the same number.

a. pretzels to crackers _____

b. crackers to pretzels _____

c. pretzels to all snacks _____

d. crackers to all snacks _____

2. Write each ratio in simplest form.

a. 12 : 18 = _____

b. 20 : 8 = _____

c. 15 : 35 = _____

3. A team has **8 girls** and **7 boys**. Jada writes **8 : 15**. What does the **15** count?

Frame *This ratio is part-to-___ because 15 counts ___.*

☐ Part-to-part

☐ Part-to-whole

15 counts_____

4. **Class Trip Vote.** A class has **24 students**. **14** vote for the museum. The **rest** vote for the aquarium.

   Step 1 Find the aquarium votes: 24 − 14.

   Step 2 Write museum : aquarium.

   Step 3 Divide both terms by the same number.

   Show your work.

   Ratio_____

5. **Find the Error.** An animal shelter has 4 cats and 10 dogs. Marcus says the ratio of dogs to cats is 2 : 5. Explain his mistake and write the correct ratio.

   Frame *Marcus wrote the ratio in the wrong ___. Dogs to cats is ___ : ___, or ___ : ___ in simplest form.*

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Count two kinds of things at home, like forks and spoons. Say the ratio both ways.

Name_____ Date_____ Period_____

LESSON 3-1 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Ratios and Tape Diagrams

Maryland Standard 6.RP.A.1 · Learning target – I can use a tape diagram to share a total according to a given ratio.

REMEMBER

- A **tape diagram** uses equal boxes; the boxes match the ratio.
- **One box** = total ÷ total parts. Multiply one box by each term. Check that the amounts add to the total.

WORKED EXAMPLE

Leah and Omar share 35 stickers in the ratio 2 : 5. Find each share.

Step 1 Total parts: 2 + 5 = 7

Step 2 One box: 35 ÷ 7 = 5 stickers

Answer **Leah: 2 × 5 = 10. Omar: 5 × 5 = 25.**

WORD BANK

tape diagram – equal boxes for a ratio

one box – total ÷ total parts

total parts – add the ratio terms

PRACTICE

show your work · label units

1. The tape diagram shows the ratio of red beads to blue beads on a bracelet. There are 48 beads in all.



Step 1 Add the boxes: 3 + 5.

Step 2 Divide 48 by the total boxes.

Step 3 Multiply one box by each term.

a. What is the value of one box? _____ beads

b. Find the red beads and the blue beads.

_____

2. Find both amounts.

a. Share 42 in the ratio 3 : 4. _____

b. Share 66 in the ratio 2 : 9. _____

3. Work Backward. A fruit cup has grapes and melon pieces in the ratio 3 : 4. It has 12 grapes.

Step 1 12 grapes are 3 boxes. Find one box.

Step 2 Multiply one box by 4 for melon.

Step 3 Add grapes and melon.

a. What is the value of one box? _____

b. How many melon pieces? _____

c. How many pieces of fruit are in the cup in all? _____

4. Stretch. Maya and Theo share stickers in the ratio **2 : 7**. Theo has **35 more** stickers than Maya.

Step 1 Theo has 7 − 2 = 5 more boxes.

Step 2 5 boxes = 35. Find one box.

a. What is the value of one box? _____ stickers

b. How many stickers do they have in all? _____ stickers

5. Find the Error. Tyrell shared 45 marbles in the ratio 4 : 5 as 36 and 45. Explain his mistake. Find the correct shares.

Frame *Tyrell divided by ___, but he should divide by the total parts, ___. Shares: ___ and ___.*

_____

_____

TEST PRACTICE

choose the best answer

6. Jamal and Rosa share $40 in the ratio 3 : 5. How much money does Rosa get?

A $5 **B** $15 **C** $24 **D** $25

7. A tape diagram shows a ratio of 4 : 3 with a total of 56. What is the value of one box?

A 7 **B** 8 **C** 14 **D** 32

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Show someone at home how to split 30 snacks in the ratio 2 : 3.